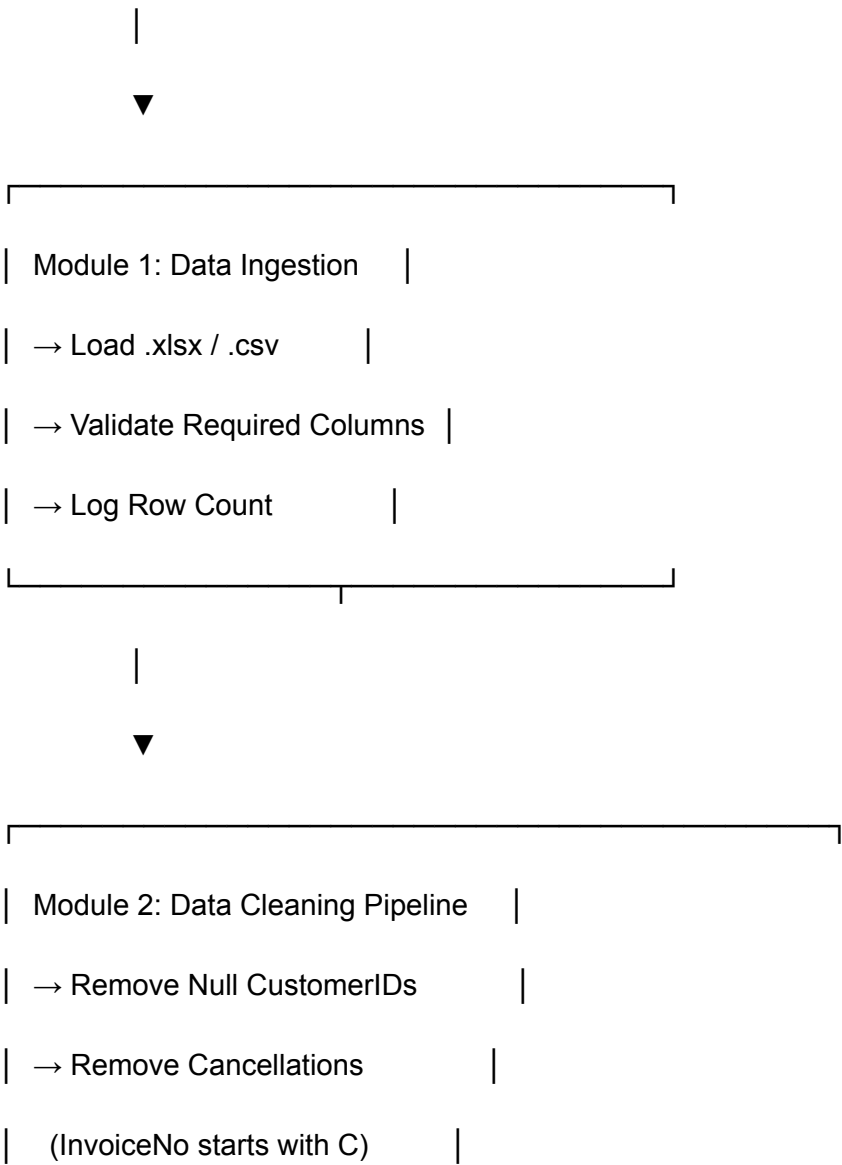


Low-Level Design (LLD)

Retail Customer Segmentation

LLD Process Flow Diagram

Start: Load Configuration (file_path, file_format)



| → Remove Negative Quantity |

| → Remove Negative UnitPrice |

| → Compute TotalPrice |

| → Treat Monetary Outliers |

| (IQR / log_cap) |

|

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| Module 3: Feature Engineering |

| → Compute RFM |

| (Recency, Frequency, Monetary) |

| → Apply log1p Transform |

| → Scale Features (MinMax / Standard) |

| → Generate Feature Matrix |

| (N_customers × 3) |

|

|



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| Module 4: Clustering Engine |

| → Fit KMeans (K=3,4,5) |

| → Store Labels + Centroids |

| → Compute Inertia (Elbow Data) |

|

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| Module 5: Model Evaluation |

| → Compute Silhouette Scores |

| → Select Best K |

| → Stability Check (Multi-Seed) |

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| Module 6: Persona Generator |

| → Compute Cluster Statistics |

| → Assign Business Personas |

| → Generate Recommendations |

| → Export personas.json |

|

|



	Module 7: API Layer (FastAPI)
	→ Load Model Artifacts on Startup
	→ /segment Endpoint
	→ /personas Endpoint
	→ /metrics Endpoint
	→ /predict Endpoint



End: Business User Consumes Segmentation Results

